

TABLE I | Two service capacities across ten CIM technologies

INT8 logical payloads · Selected reference configurations · Sustained complete-matrix loading

Technology / K × N	Clock	Streaming	Resident	Service ratio	Crossover	Availability
	P [ns]	ρ [MB/s]	τ [MB/s]	RI^* [-]	U^* [-]	a [-]
 SRAM ACIM 128×128	5.0	44	3,200	0.014	1.8	1.0
 SRAM DCIM 128×16	5.0	390	3,200	0.12	1.9	1.0
 2D NOR 128×128	8.5	20	0.080	260	33,000	1.0
 3D NAND 4608×240	9.0	6.7	0.58	12	2,800	1.0
 RRAM 128×64	5.0	29	4.3	6.7	430	1.0
 MRAM 256×32	9.0	180	63	2.8	88	1.0
 PCM 256×128	5.0	2.8	7.0	0.40	51	1.0
 HfO₂ FeRAM 128×128	8.5	22	99	0.23	29	1.0
 Gain-cell eDRAM 64×64	5.5	14	160	0.089	5.7	0.83
 3D FeFET 128×128	8.5	36	54	0.67	86	1.0

$\rho = B_S/\Delta_S$, $\tau = B_R/T_R$, $RI^* = \rho/\tau$; $U^* = T_R/\Delta_S = NRI^*$ for the selected one-Byte input and weight payloads.

Both capacities use complete service boundaries. Gain-cell is volatile; its rates and effective intervals include periodic refresh.

Internal planes, reference pages, verification and refresh consume resources and time without increasing logical payload.

Native sizes, precision conditions and resources differ; these capacities do not rank equal-area or equal-work implementations.

Device services combine literature estimates with NeuroSim peripheral models. 3D FeFET denotes vertical AND FeFET; see METHOD for scope.

Typical streaming and resident capacity

Selected reference INT8 payloads · $RI^* = \rho/\tau$



Native sizes and resources differ; these capacities do not rank equal-area or equal-work implementations.